Understanding or Imitation? Auditing Conceptual Understanding and Reasoning in Large Language Models

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Abstract

Do large language models genuinely understand concepts, or do they merely reproduce patterns that resemble understanding? As benchmarks increasingly serve as proxies for conceptual competence, it is critical to examine whether high performance reflects coherent reasoning or superficial imitation. We audit benchmark-based claims about conceptual understanding in LLMs through a large-scale reproducibility and extension effort conducted in the context of "Potemkin Understanding in Large Language Models" study. We replicate evaluation pipelines that test whether models can define concepts and apply them consistently across tasks. While we confirm that many standard non-reasoning models exhibit gaps between definition accuracy and downstream application, we find that these effects are substantially less stable and more methodologically fragile than reported.

Across five identical runs per model, the reported potemkin scores vary by as much as 31% - large enough to reverse qualitative conclusions - revealing significant stochastic instability driven by undocumented randomness and pipeline assumptions. We identify an implicit capability threshold in automated procedures: smaller models frequently fail to generate coherent sub-questions, causing silent breakdowns or misleading scores. Conditioning evaluation on correct definitions risks conflating memorized recall with genuine conceptual understanding. In addition, the testing pipeline is prompt-dependent and relies on same-model self-judging, which can mask consistent but incorrect interpretations and inflate apparent coherence. We further show that domain aggregation masks large reliability differences across auto-gradable and human-annotated domains. Most strikingly, reasoning-focused models reduce incoherence by an order of magnitude, suggesting that apparent conceptual failures depend strongly on inference-time regime rather than model class. Our findings indicate that claims about widespread conceptual incoherence in LLMs are directionally supported but empirically fragile. We provide a reproducibility map and concrete recommendations for building more stable and interpretable evaluations of model understanding.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

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Overview of Reproducibility Efforts and Contributions

This paper is written and structured to be an experience paper.
We document what is reproducible, what is fragile, what surprises us, and what the community should do differently. The anchor paper (Manoridis et al. 2025) provides code and data for some components; others required reconstruction or extension to the original code repository. We contribute to building a clean, complete reproducibility map, extended experiments to newer models, and notng down lessons that generalise beyond this specific paper.

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1 Introduction

1.1 The Stakes of LLM Evaluation

Large language models are now embedded in high-stakes decision pipelines across medicine, law, education, and software engineering. In each of these contexts, we are implicitly trusting that when a model produces a correct-sounding answer, it is doing so for the right reasons i.e that it actually understands the underlying concept, not just the surface pattern of how that concept tends to be tested. This assumption has gone largely unexamined. Factual errors (or otherwise known as hallucinations) in LLM outputs can be caught by cross-referencing known facts, but conceptual hallucinations - where a model uses a term fluently yet applies the underlying concept incoherently - are far harder to detect and far more dangerous to rely upon.

This trust is largely mediated through benchmarks. Performance on datasets like MMLU, BIG-Bench, and HumanEval has become the de facto language for claims about LLM capability and reliability. A model that scores 90% on a professional medical exam is implicitly credited with something like "medical understanding." But this inferential leap, from benchmark score to genuine conceptual grasp, rests entirely on an assumption that is almost never stated explicitly, let alone verified.

This tension is formalised by [9] (hereafter referred to as: *the Potemkin paper*), which introduces the concept of a keystone set and a measurable potemkin rate. A Potemkin model is one that passes definition-level tests but fails systematically at downstream concept application tasks exhibiting the appearance of understanding without substance. The paper reports that potemkin behaviour is ubiquitous across all models.

These are strong claims with significant implications for the entire field of LLM evaluation. If true, they invalidate a substantial fraction of existing benchmark results. If overstated or methodologically fragile, they risk producing an incorrect picture of model capabilities. The reproducibility of the specific empirical findings is one question; the robustness of the underlying methodology is another; and the generalizability of the results to models released after the study is a third. Our paper is a large-scale reproducibility and extension effort that attempts to address all three targeting the Potemkin paper's two experimental procedures.

1.2 Contributions

This paper is organized around four research questions:

- **RQ1:** Can the original empirical results be reproduced using the provided code and data?
- **RQ2:** What components required reconstruction, and what did that process reveal about the original methodology?
- **RQ3:** Do potemkin rates hold, increase, or decrease on models untested or released after the original study?
- **RQ4:** What are the conceptual and methodological limitations of the potemkin framework as a diagnostic tool?

Our contribution is threefold:

- **Reproduction:** We attempt exact reproduction of the automated lower-bound procedure (Procedure 2) and partial reproduction of the benchmark procedure (Procedure 1).
- **Extension:** We extend both procedures to models not included in the original study, including smaller open-source models, newer reasoning models, and quantised variants.
- **Critique:** We identify and systematically test four sources of methodological fragility: stochastic instability across runs, capability-threshold failures in smaller models, same-model self-judging circularity, and keystone validity.

2 BACKGROUND AND RELATED WORK

2.1 The Potemkin Paper: Technical Summary

The Potemkin Paper [9] makes a single core claim: Standard human-designed benchmarks are invalid tests for LLMs because LLMs can misunderstand concepts in non-human ways. The argument rests on the notion of a *keystone set* — a minimal set of instances that, if answered correctly by a human, implies correct concept understanding. An LLM that passes a keystone but fails on related use tasks is defined as exhibiting *potemkin understanding*.

Two empirical procedures are introduced:

- **Procedure 1 (Benchmark):** A custom benchmark across 32 concepts in 3 domains — literary techniques, game theory, and psychological biases — testing definition, classification, generation, and editing tasks. The potemkin rate is defined as the fraction of use-task questions answered incorrectly, conditional on a correct definition.
- **Procedure 2 (Automated):** A fully automated lower-bound procedure using MMLU: the model answers benchmark questions, generates related subquestions, answers them, then judges its own answers. Disagreement between benchmark performance and subquestion performance indicates potemkin behaviour.

Key reported results: mean potemkin rates of 0.55 (classify), 0.40 (generate), and 0.40 (edit); automated lower bound of 0.62; and a striking outlier: *o3-mini* achieves 0.03 incoherence versus GPT-4o's 0.64.

2.2 Related Work

The call to closely analyze the ability of machine-learning trained systems to reason correctly has been a concern for years [3]. With the rise of highly performant LLMs displaying human-like abilities to reason across a broad of tasks and environments, there has been growth in works scrutinzing and questioning the ability of current generation LLMs to actually reason [2]. This is of particular concern in the area of AI alignment, where systems may need to perform complex inferences to determine if their behavior in what amounts to open-world domains matches meets expectations and stays within safety constraints [1].

Chief amongst many concerns is the tendency of machine learning systems to learn *shortcuts*, surface feature cues in the input that directly correlate with the answer. Shortcuts gives the impression the system is capable of complex conceptual understanding, but is limited in practice by the domain of the shortcut and thus does not generalize. Shortcutting has been a long running concern for machine learning systems [3], especially for systems performing natural language understanding (NLU) tasks [4]. NLU systems are expected to perform complex reasoning over inputs and generate

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Property	Procedure 1 (Benchmark)	Procedure 2 (Automated)
Scope	3 domains, 32 concepts	Any MMLU / BBH concept
Human annotation required?	Yes (significant expert effort)	No
Bound type	Direct measurement	Lower bound
Code available?	Partial	Full
Scalable to new models?	No, expert-intensive	Yes
Judge bias risk?	Author/expert labels	Same-model self-judging

Table 1: Comparison of reproducibility characteristics across the two Potemkin procedures

answers in natural language, such as answering natural language questions, and the area has driven much of the benchmarks used to assess LLMs [5, 11, 14]. In particular, the consistency and coherence of question-answering systems have been examined under conceptual variance [12] and logically consistency [12].

Prior work has measured the ability of LLMs to perform reasoning against hierarchical models of task complexity, most notably Bloom's Taxonomy [8], a framework originally developed for pedagogy where problems are mapped into a hierarchy of increasingly difficult problem types organized around the type of cognitive ability required. Indeed, Bloom-based analyses of LLM performance on question-answer benchmarks [7] and medical examination [6] show these systems tend to perform poorly on higher order reasoning.

In the area of education research concept inventories, sets of questions that measure the level of concept understanding displayed by students and are equivalent to the keystone set described in [9], have found use in assessing student learning across multiple domains [13]. However, there is significant variance in the development, validation, and reliability of such inventories [10]. Indeed establishing psychometric validity and reliability (measurement of mental processes) is critical for education and is central to our recommendations for assessments of LLMs and other automated systems: If an instrument fails to generate reliable measures of a phenomena in question, it compromises the ability to make concrete, trustworthy statements about that phenomena.

3 REPRODUCIBILITY ATTEMPT: SETUP AND METHODOLOGY

This section provides the complete specification of our reproduction and extension efforts, detailed to the level required for independent replication.

To our knowledge, no published reproducibility study has targeted the Potemkin paper prior to this work. The paper was accepted at ICML 2025; code was made available via GitHub at commit 39 of the `PotemkinBenchmark` repository. Two procedures were employed in the work to assess LLM understanding (Table 1). Procedure 1's annotation pipeline relies on expert annotators recruited through Upwork and is not independently replicable without significant cost. Procedure 2's code is available and fully automated, making it the primary target for reproduction.

Component	Specification
Operating System	Ubuntu 24.04.4 LTS
CPU	Intel Core i9-14900K (32 threads)
RAM	64.0 GiB
Storage	2 TB SSD
GPU	2 × NVIDIA Quadro RTX 6000 (48 GB VRAM each)
Python	3.10 (exact version documented in `requirements.txt`)
Primary libraries	`transformers`, `torch`, `openai`, `anthropic`, `datasets`
Artifact commit	`PotemkinBenchmark` @ commit 39 (GitHub)

Table 2: Hardware and software environment used for reproduction and extension experiments

3.1 Experimental Environment

3.2 Hardware and Software Environment

All reproduction and extension experiments were conducted on a Linux workstation equipped with dual RTX 6000 GPUs. The software environment used PyTorch together with the Hugging Face `transformers` library for local inference and API-based model interaction.

The original experiments primarily relied on API-accessed models and therefore did not require local GPU inference. GPU execution was introduced not only for extension experiments involving smaller open-weight models, including `Llama-3.1-8B`, `Llama-3.2-3B`, and selected `Qwen-2.5` variants but also the original open source models.

3.3 Artifact Acquisition and Assessment

We cloned the `PotemkinBenchmark` repository at commit 39, the version current at the time of paper acceptance, and verified the SHA against the authors' published hash. Upon acquisition, we assessed the available artifacts against the procedures described in the paper.

What was directly usable. The repository includes the game theory domain data together with its automated evaluation functions, the MMLU/BBH-based automated lower-bound procedure (Procedure 2) with its full pipeline code, the raw benchmark datasets for classification, generation, and editing tasks across all three domains, and the prompt templates used for definition elicitation and task evaluation. Procedure 2 code was executed without modification as an initial baseline to establish reference values prior to any extension.

What required reconstruction or extension. No complete execution pipeline is provided for Procedure 1 (benchmark evaluation). Although the paper reports results across seven models, 32 concepts, and four task types, the repository contains only the underlying data and isolated evaluation functions, without a unified runner

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capable of reproducing Table 1 directly. The experiment orchestration layer was therefore reconstructed from the methodological description in the paper, treating the published question sets as inputs and implementing automated evaluation for the game theory domain only, where auto-grading functions are defined.

A further limitation concerns the annotation-dependent domains. The psychological biases annotations were produced by behavioural scientists recruited through Upwork and are not reproducible in a strict methodological sense: the annotators are not replaceable, their individual judgments are not published, and no inter-rater reliability statistics are reported. We treat this as a hard reproducibility boundary for that domain. The literary techniques annotations were produced by the paper's own authors without independent external validation, a limitation revisited in Section 5.

Code quality observations. The released code is functional but uneven in quality. The automated procedure is runnable, but includes an undocumented subquestion generation cap that directly affects reported potemkin rates (discussed in Sections `"""`).

3.4 Modifications Required for Reproducibility

The following modifications were required before stable reproduction could be achieved.

Procedure 2 (Automated Framework). The original pipeline caps subquestion generation at exactly five: if a model generates more, the pipeline re-runs the entire generation call, creating unnecessary computational cost and introducing additional non-determinism. We modified this behaviour by randomly sampling five questions whenever more than five were produced.

Smaller models frequently generated fewer than five subquestions on the first attempt. To address this, we introduced a retry loop that re-prompts up to three times before falling back to the available output, while explicitly logging any shortfall.

The `relies_on_concept` filter also failed silently for smaller models that included the token `YES` inside otherwise incoherent responses. We added explicit parsing rules to detect these cases and record them as evaluation failures rather than valid responses.

We additionally implemented automated saving of per-run potemkin rate scores to structured CSV output, which was absent from the original release, and extended the pipeline to support local inference through Hugging Face `transformers`, enabling evaluation of open-source and quantised models without external API cost.

Procedure 1 (Benchmark Framework). No executable code was provided for Procedure 1. We therefore reconstructed the evaluation pipeline from scratch following the methodology described in the paper and applied it to the published question sets.

3.5 Experimental Design

3.5.1 Stability Analysis. To assess stochastic instability, we ran Procedure 2 five times with identical configurations for each model, using the same dataset sample and hyperparameters. We report the mean and standard deviation of the potemkin rate across runs. The datasets used were MMLU and BigBench Hard (BBH). This tests whether the single-run values reported in the original paper are representative.

3.5.2 Capability Threshold Analysis. We extended Procedure 2 to models with parameter counts ranging from 0.5B to 70B: `Llama-3.1-8B-Instruct`, `Llama-3.2-3B-Instruct`, `Qwen-2.5-0.5B`, `Qwen-2.5-7B`, and `Llama-3.3-70B` (4-bit quantised). We report not only potemkin rates but also pipeline failure modes, specifically cases in which models fail to generate valid subquestions, since these represent a form of evaluation breakdown not discussed in the original paper.

3.5.3 Cross-Model Judging Experiment. The original Procedure 2 uses the same model as both responder and judge, creating a circularity risk: a model that consistently applies an incorrect concept may still appear coherent under self-judging. We decouple these roles by running a 3×3 cross-judging matrix in which three responder models are evaluated by three different model families, and compare the resulting potemkin rates to the self-judging baseline. This constitutes the most novel methodological contribution of the present study.

3.5.4 Reasoning vs. Non-Reasoning Model Extension. The original paper includes `o3-mini` and `GPT-o1-mini` in its automated procedure and reports substantially lower incoherence scores (`o3-mini`: 0.03 versus `GPT-4o`: 0.64), but does not provide a dedicated analysis of this contrast. We extend the comparison to include `DeepSeek-R1-Distill-Qwen` and `gpt-oss`, and run the full analysis across both MMLU and BBH. The goal is to determine whether reasoning-mode inference is the key explanatory variable, or whether the effect is confounded with model scale.

3.5.5 Alternative Keystone Experiment (Priority 5). The benchmark framework treats correct definition generation as the keystone indicator of concept understanding against which downstream failures are measured. If definition tasks are especially susceptible to surface-pattern recall, reported potemkin rates may be inflated. For a subset of game theory concepts, we therefore replace the definition keystone with two alternatives: (a) a multiple-choice recognition task, and (b) a classification-only keystone. We then compare resulting potemkin rates to assess whether keystone choice materially drives the reported effect.

The clearest evidence of the reproducibility problem comes from our five-iteration runs across models and datasets. Rather than the stable point estimates reported in Table 2 of the paper, we find substantial variance across identical runs:

Model	Dataset	Iter 1	Iter 2	Iter 3	Iter 4
Llama-3.1-8B-Instruct	MMLU	1.11 (0.04)	1.05 (0.02)	0.90 (0.04)	0.80 (0.03)
Llama-3.1-8B-Instruct	BBH	1.09 (0.03)	1.14 (0.04)	0.90 (0.03)	1.09 (0.04)
Llama-3.2-3B-Instruct	MMLU	1.08 (0.02)	1.04 (0.02)	1.10 (0.04)	0.91 (0.02)
Llama-3.2-3B-Instruct	BBH	1.06 (0.03)	0.82 (0.03)	1.01 (0.07)	0.87 (0.07)
Qwen-2.5-0.5B	MMLU	0.90 (0.13)	1.44 (0.09)	—	1.03 (0.03)
Qwen-2.5-0.5B	BBH	ERROR	1.75 (0.12)	0.78 (0.09)	ERROR
Qwen-2.5-7B	MMLU	0.79 (0.01)	1.05 (0.03)	0.74 (0.02)	0.81 (0.01)
Qwen-2.5-7B	BBH	0.72 (0.01)	0.80 (0.02)	0.84 (0.03)	0.73 (0.02)

Table 3: Performance across iterations for different models on MMLU and BBH benchmarks.

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The range of incoherence scores for Llama-3.1-8B on MMLU alone spans from 0.80 to 1.11 across five identical runs, a swing of 0.31 on a scale where 1.0 represents chance performance. For Qwen-2.5-7B on MMLU, the range is 0.74 to 1.05. These are not minor fluctuations around a stable mean; they are differences large enough to change the qualitative interpretation of whether a model is performing above or below chance. The paper reports single point estimates with standard errors computed across concepts, but makes no acknowledgment of run-to-run variance of this magnitude. For smaller models, the procedure breaks down more severely. Qwen-2.5-0.5B failed to generate any sub-questions at all in two MMLU runs and produced a NoneType parsing error in three BBH runs. This exposes a structural assumption baked into the automated procedure: it presupposes that the model is capable of reliably generating coherent sub-questions from MMLU-style inputs. Models below a certain capability threshold cannot satisfy this precondition, which means the procedure cannot be applied to smaller or weaker models without modification. The original paper restricts itself to seven large frontier models and never acknowledges this limitation.

3.6 Methodological Issues

The most consequential methodological weakness is what we call the keystone assumption: the entire Procedure 1 benchmark is conditioned on models correctly defining a concept, treating definition accuracy as the keystone that validates subsequent task evaluation. But the paper's own data undermines this choice: models define concepts correctly 94.2% of the time. This is a suspiciously high rate. LLMs have been trained on enormous corpora of definition-question pairs, and producing a correct definition may be precisely the task most susceptible to surface-level pattern matching. A model that has encountered the phrase "a haiku is a three-line poem with a 5-7-5 syllabic structure" thousands of times during pretraining can reproduce it without any coherent internal representation of syllabic structure. This matters because the potemkin rate is computed conditional on correct definition. If the definition keystone is trivially easy to pass - not because models understand concepts, but because they have memorized definitions - then the reported potemkin rate is not measuring a gap between understanding and application. It is measuring a gap between an easily-faked signal (definition recall) and a harder task (concept application). The framework would then be documenting a difficulty asymmetry, not a conceptual incoherence. A second methodological concern is that the automated procedure (Section 4) uses the same model as both responder and judge. A model that consistently misapplies a concept but does so in a self-consistent way will appear coherent under self-judging. This conflates two distinct failure modes: incoherence (the model contradicts itself) and misunderstanding (the model applies a consistently wrong interpretation). The paper describes these as separate phenomena but the automated procedure cannot distinguish them when the same model plays both roles. Domain aggregation also obscures important signal. Game theory concepts are formally specified and auto-graded; literary techniques are subjective and annotated by paper authors; psychological biases require expert behavioral scientists to evaluate. Aggregating potemkin rates across these three domains, as the paper does in Table 1, blends results with very different reliability levels. Table 6 in the paper's appendix shows enormous variance across domains and concepts, but this variance receives no dedicated analysis.

4 LESSONS FOR THE COMMUNITY

The Potemkin paper makes a genuine and important contribution to how the field thinks about benchmark validity. The theoretical framework is rigorous, the core empirical finding is directionally robust, and the choice to release data and code publicly reflects commendable transparency. The following lessons are offered not as criticism of the paper's conclusions, but as specific, actionable improvements that would make the contribution more durable and replicable. The most urgent improvement is documentation of run-to-run variance. The paper reports point estimates with standard errors computed across concepts, but these standard errors reflect cross-concept variation, not the stochastic instability of the procedure itself. Our five-iteration replication shows that a single run of the automated procedure on the same model and dataset can yield incoherence scores ranging by 0.20 to 0.30 — a range large enough to change the qualitative interpretation of results. Reporting variance across runs, and providing seeded-random versions of the pipeline, would substantially increase confidence in the reported numbers. The subquestion generation cap should be documented explicitly, its value reported in the paper, and its effect on results analyzed. It is currently an invisible parameter that shapes the denominator of the incoherence rate without acknowledgment. The definition-evaluation procedure for Procedure 1 was performed by the paper's own authors and reports 94.2% definition correctness, but no inter-rater reliability statistics are provided. Future versions should have a second annotator evaluate at least a stratified sample of definitions and report Cohen's kappa. The conditioning variable for the entire benchmark analysis deserves the same scrutiny as the outcome variables. The reasoning model finding belongs in the main paper, not buried in Table 2. An order-of-magnitude reduction in incoherence between standard and reasoning models is the most practically significant result in the paper. It deserves its own section, its own analysis, and given the pace of reasoning model deployment, its own careful discussion of what it implies for the framework's claims about ubiquity.

4.1 For the Broader Research Community

Our experience reproducing this paper points to several systemic lessons that generalize well beyond this specific work. Evaluation papers are held to lower reproducibility standards than they should be. A paper that introduces a new benchmark or evaluation procedure is, implicitly, building infrastructure that downstream work will rely on. If that infrastructure is fragile — if results vary widely across runs, if key parameters are undocumented, if the annotation pipeline cannot be replicated — then every paper that cites it and builds on it inherits that fragility. The reproducibility bar for evaluation methodology papers should arguably be higher than for standard empirical papers, not lower. At minimum, benchmark papers should be required to report run-to-run variance, document all stochastic parameters and seeds, and provide a minimal end-to-end reproduction script that a reader can run without reconstructing the orchestration layer. Domain aggregation conceals more than it

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reveals. The aggregate potemkin rate in Table 1 mixes results from a formally specified auto-gradable domain with results from interpretive, expert-annotated domains. The variance across domains visible in Appendix Table 6 of the original paper is large enough to tell very different stories depending on which domain you examine. Future work in this area should resist the temptation to produce a single headline number and instead report domain-stratified results as the primary finding, with aggregates treated as secondary summaries. The self-judge design is a structural limitation that should be acknowledged by default. Using a model to evaluate its own outputs is expedient and, as the original paper argues, conservative in principle — a self-consistent but wrong model will appear coherent. But our cross-model judging experiments suggest that this conservatism may be purchasing convenience at the cost of validity. When a model consistently applies a wrong interpretation that it also consistently judges to be correct, the self-judging procedure fails to surface the misunderstanding entirely. The field should treat same-model evaluation as a baseline that requires cross-model validation before results are accepted. Reasoning models change the conclusion, not just the numbers. The finding that reasoning models achieve dramatically lower incoherence rates than standard models is not a quantitative refinement of the paper's central claim. it is a potential refutation of its universality. If the potemkin failure mode is substantially mitigated by chain-of-thought reasoning, then the claim that potemkins are ubiquitous across LLMs may be a claim about a specific inference-time regime rather than about LLMs as a class. As reasoning-mode inference becomes more prevalent in deployment, this qualification becomes increasingly important. The community should rerun all potemkin analyses on reasoning models before treating the ubiquity claim as settled.

5 CONCLUSION

This paper has attempted to do three things: reproduce the core empirical claims of the Potemkin Understanding paper, characterize the fragilities and failures encountered in that effort, and extend the evaluation to model families and conditions not covered by the original study. Our reproduction confirms the paper's directional finding: standard non-reasoning LLMs do exhibit high incoherence rates under the automated procedure, and the gap between definition accuracy and task application accuracy in the benchmark is real. These results are not fabricated and the framework is not wrong. But the specific numbers are not as stable as the paper's presentation suggests. Run-to-run variance in the automated procedure is large enough to materially affect conclusions for individual models. The procedure fails for smaller models in ways that are not acknowledged and not easily corrected. The benchmark's aggregate potemkin rates inherit unknown measurement error from unvalidated annotation pipelines. And the single most striking finding in the paper, the order-of-magnitude reduction in incoherence for reasoning models, is left unanalyzed despite being the result most consequential for how the field should interpret the paper's central claim.

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A Research Methods

A.1 Part One

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A.2 Part Two

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B Online Resources

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